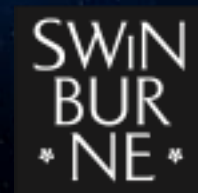


Discovering the Origins of Gravitational-Wave Sources

Mid-Candidature Review

Ashwathi Nair

Primary Supervisor : Dr. Simon Stevenson
Secondary Supervisor : Prof. Jarrod Hurley



Centre for
Astrophysics and
Supercomputing

Discovering the Origins of Gravitational-Wave Sources

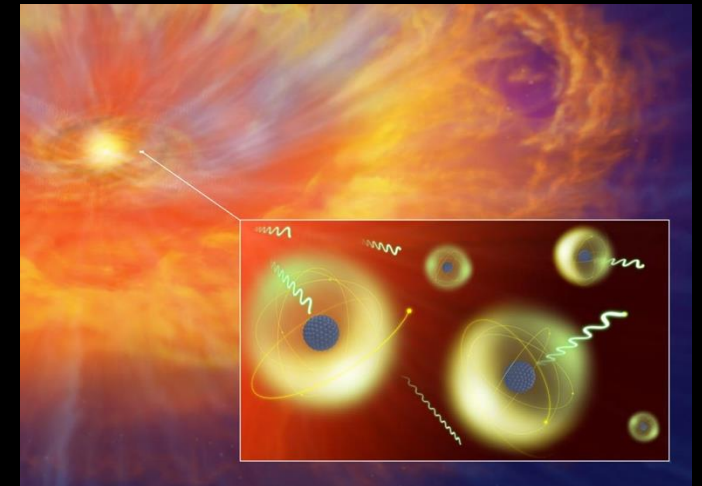
Run	Dates	Duration (months)	Detections
O1	September 18, 2015 to January 12, 2016	3	3
O2	November 30, 2016 to August 25, 2017	9	8
O3	April 1, 2019 to March 26, 2020	11	79
			Candidates
O4	May 24, 2023 to March 19, 2025	23	200

O1 to O3 included about 23 months-worth of observing time between 2015 and 2020. In that time, 90 GW detections were made. In the first 23 months of O4, 200 GW candidates have been logged. O4 is currently scheduled to end in November, 2025.

Why care about double neutron stars?



- Supernovae explosion (SNe)
- Gamma ray burst (GRB) and Kilonova
- Formation of heavy r-process elements
- Binary stellar evolution



What do we know about DNS systems so far ?

Detected by radio telescopes and gravitational-wave detectors

Radio Astronomy

of ~ 26 Galactic DNS in the field between total mass
2.50–2.89 $M_{\odot}$

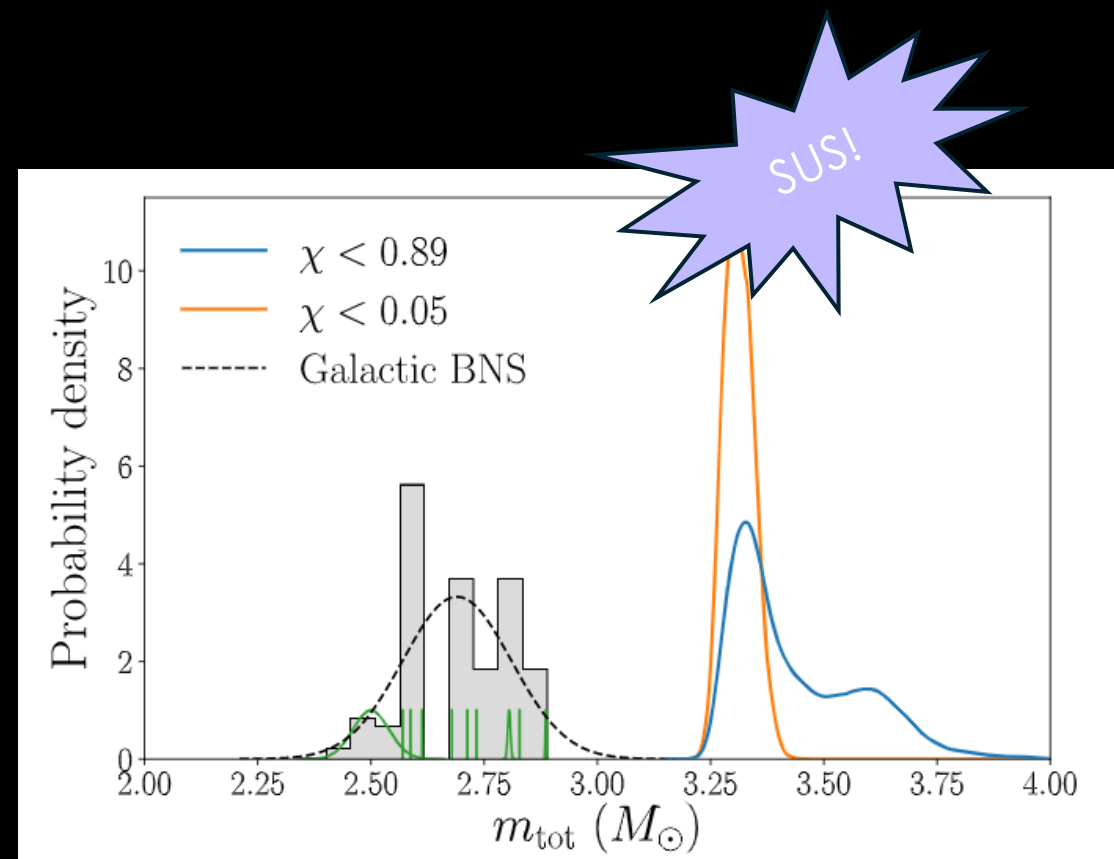
Gravitational Waves

GW170817

Total mass : $2.87^{+0.47}_{-0.09} M_{\odot}$

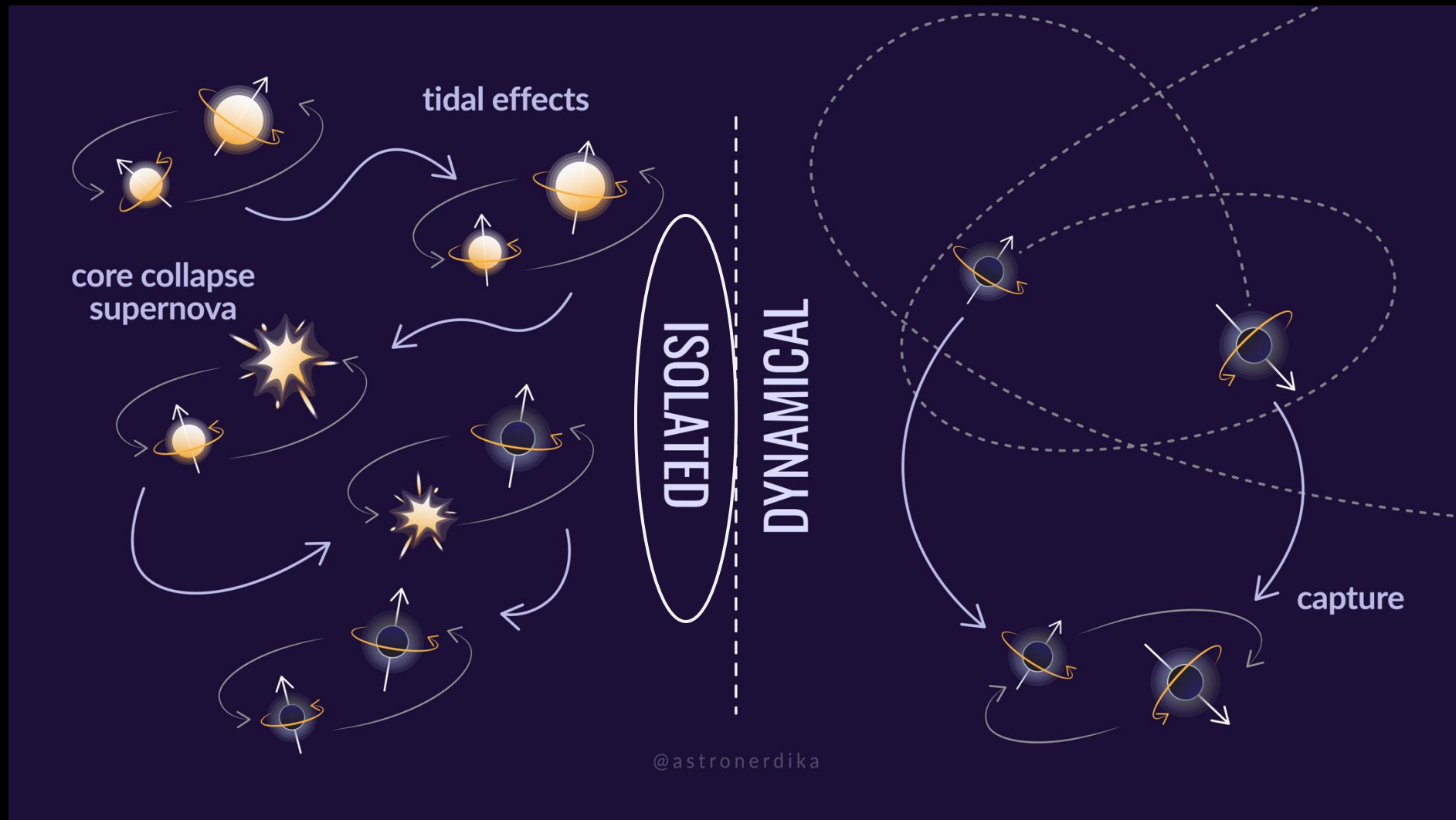
GW190425

Total mass : $3.4^{+0.3}_{-0.1} M_{\odot}$

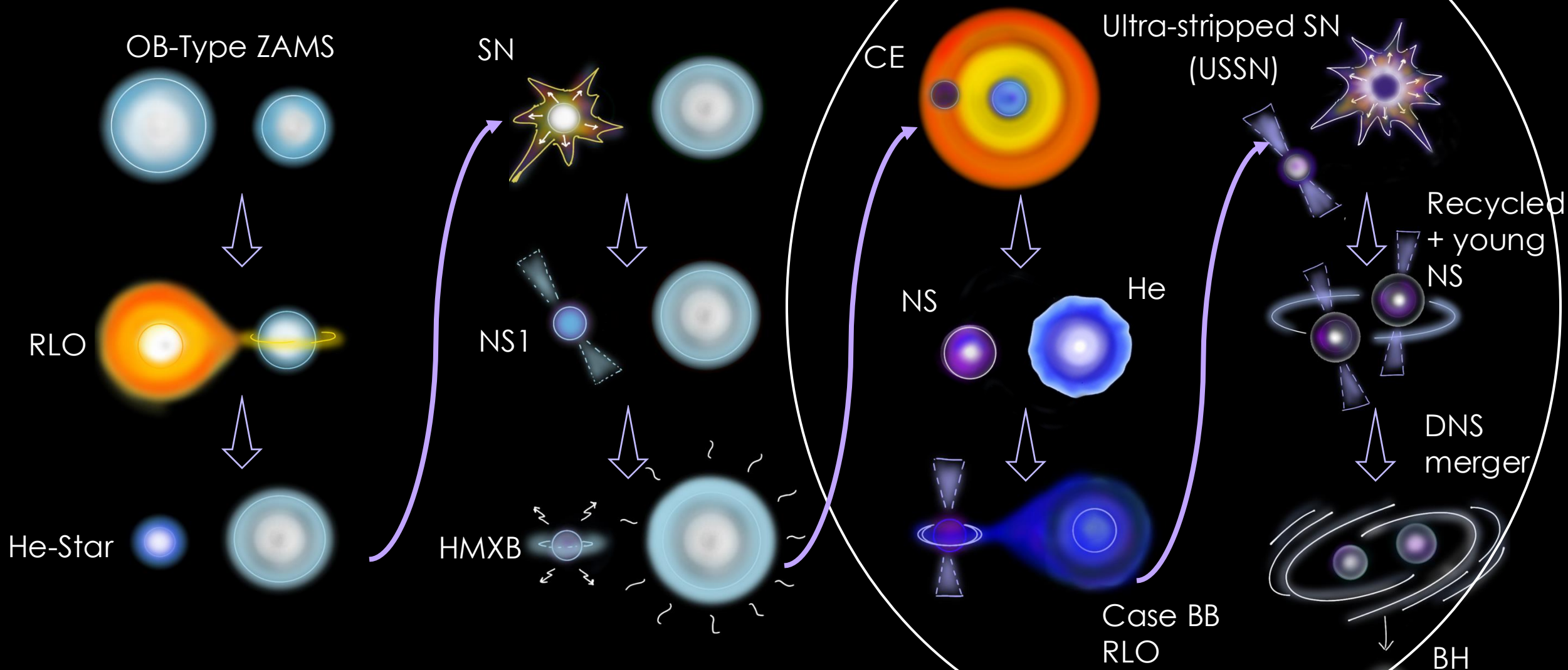


Abbot et al. 2020

Formation Channels of Compact Binary Mergers

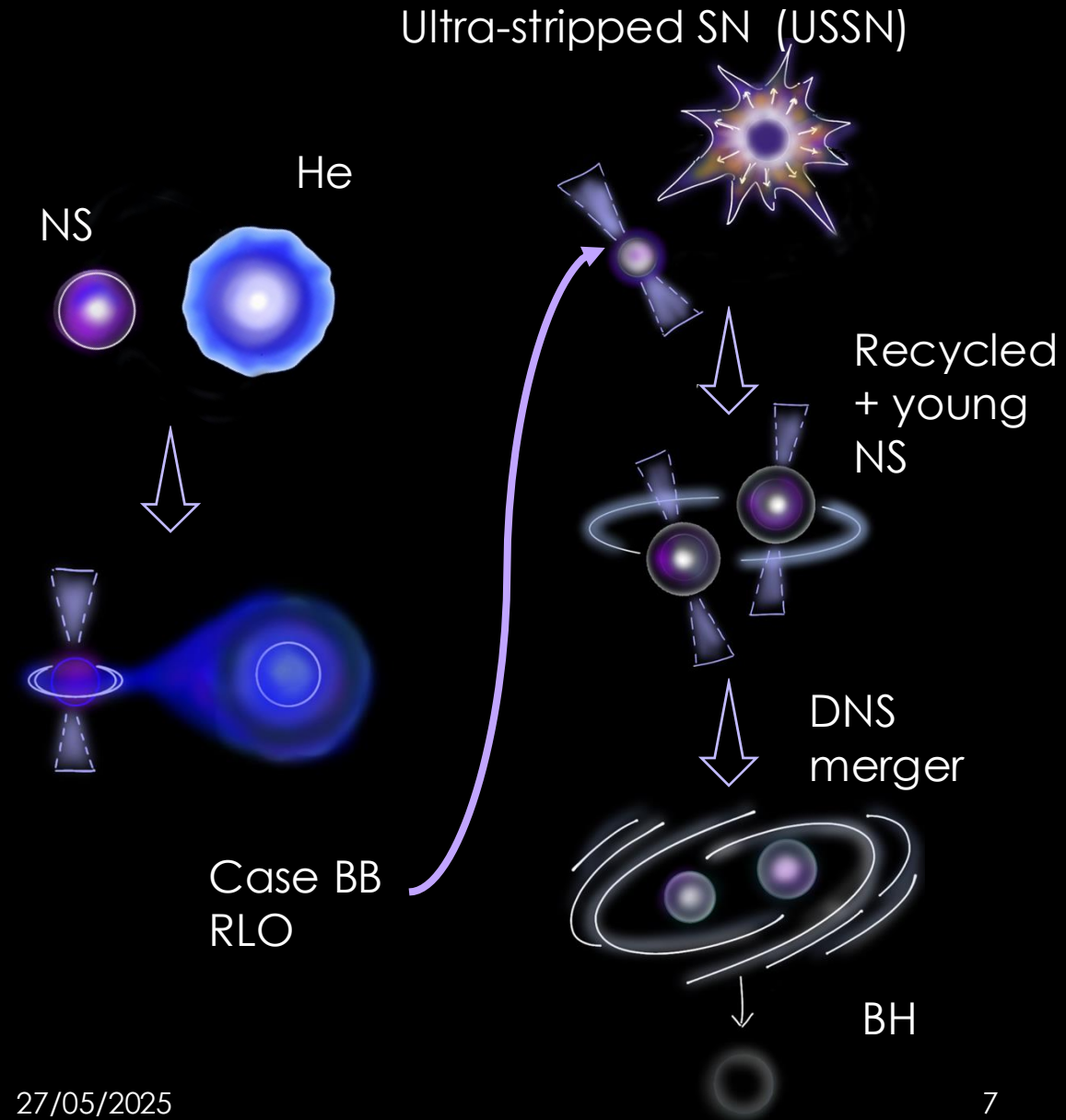


Standard Double Neutron Star Formation Pathway – T. Tauris et al. 2017

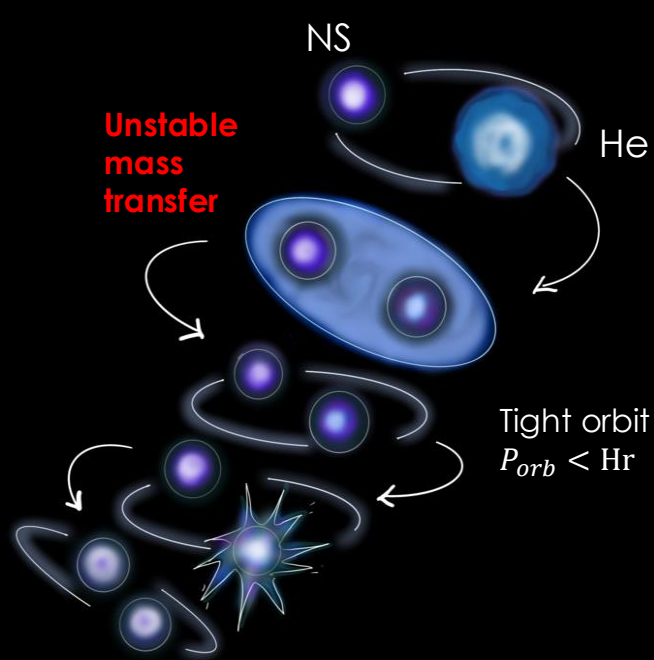


The evolutionary phases that decides the fate of DNSs

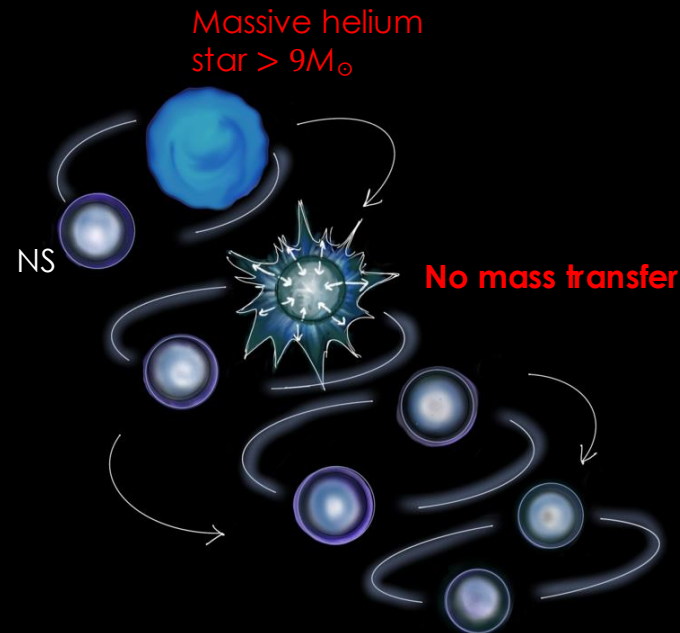
Preferred for modeling DNS formation pathway



Alternate Proposed Formation Pathways for Massive Double Neutron Star



B.P. Abbott et al.
(2020)



Vigna Gómez et al.
(2021)

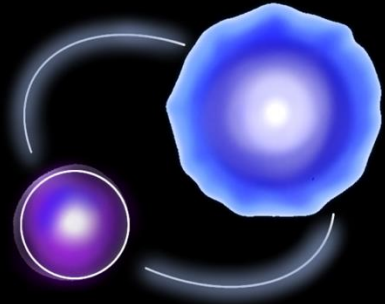
Other uncertainties:

- **Formation at lower metallicities** - Giacobbo & Mapelli (2018)
- **Super-Eddington accretion** - Zhang et al. (2023)
- **Case BB RLO MT with a first-born NS** - Qin et al (2024)

No self-consistent unbiased formation pathway for heavy double neutron stars

Project I, Part A

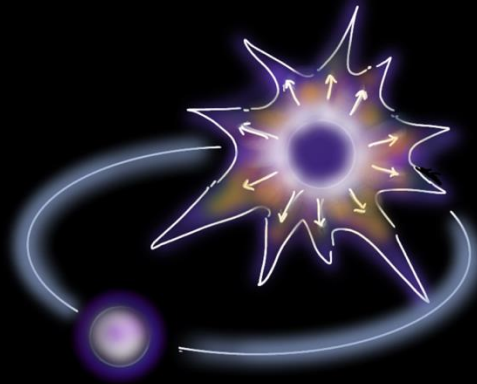
Framework Created for the Formation Pathway



ZAMS He (at solar metallicity)-
 $1.4 M_{\odot}$ NS binary evolution

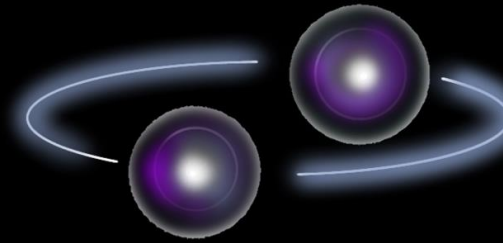
MESA

Paxton et al. 2011



Incorporating supernova and
kick prescription

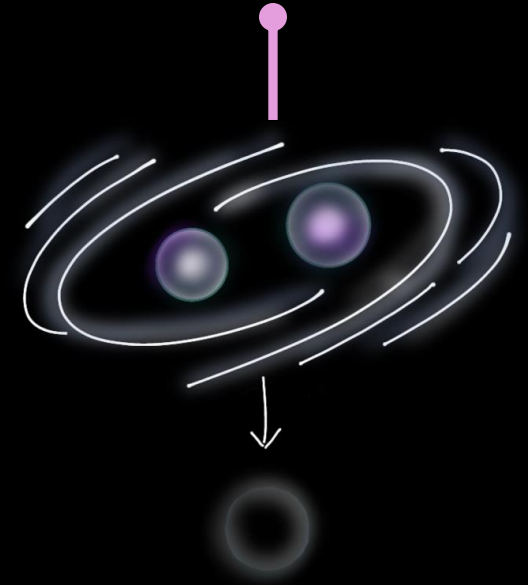
Mandel & Müller 2020,
Kapil et al. 2023



Post-Supernova effects on
orbital dynamics

Brandt &
Podsiadlowski 1995

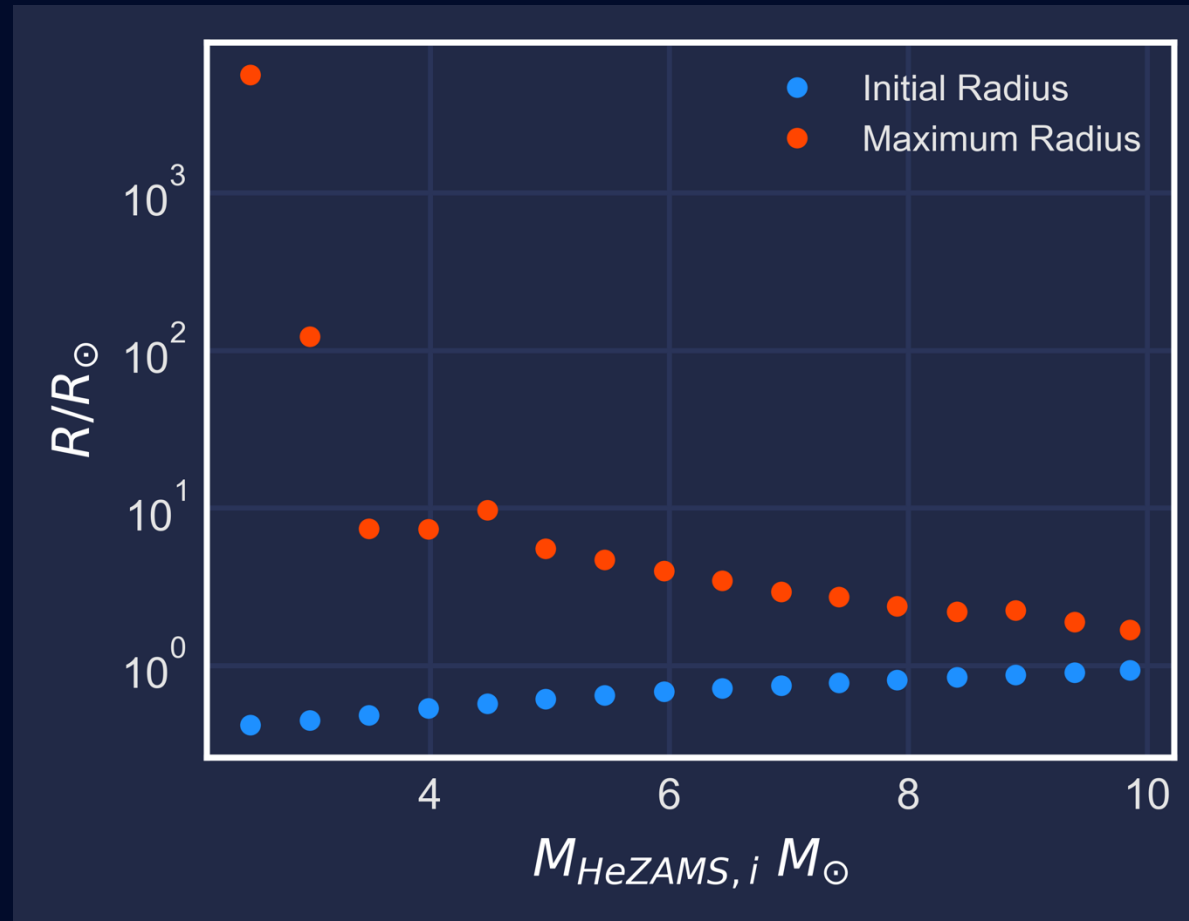
Gravitational-wave
merger timescale



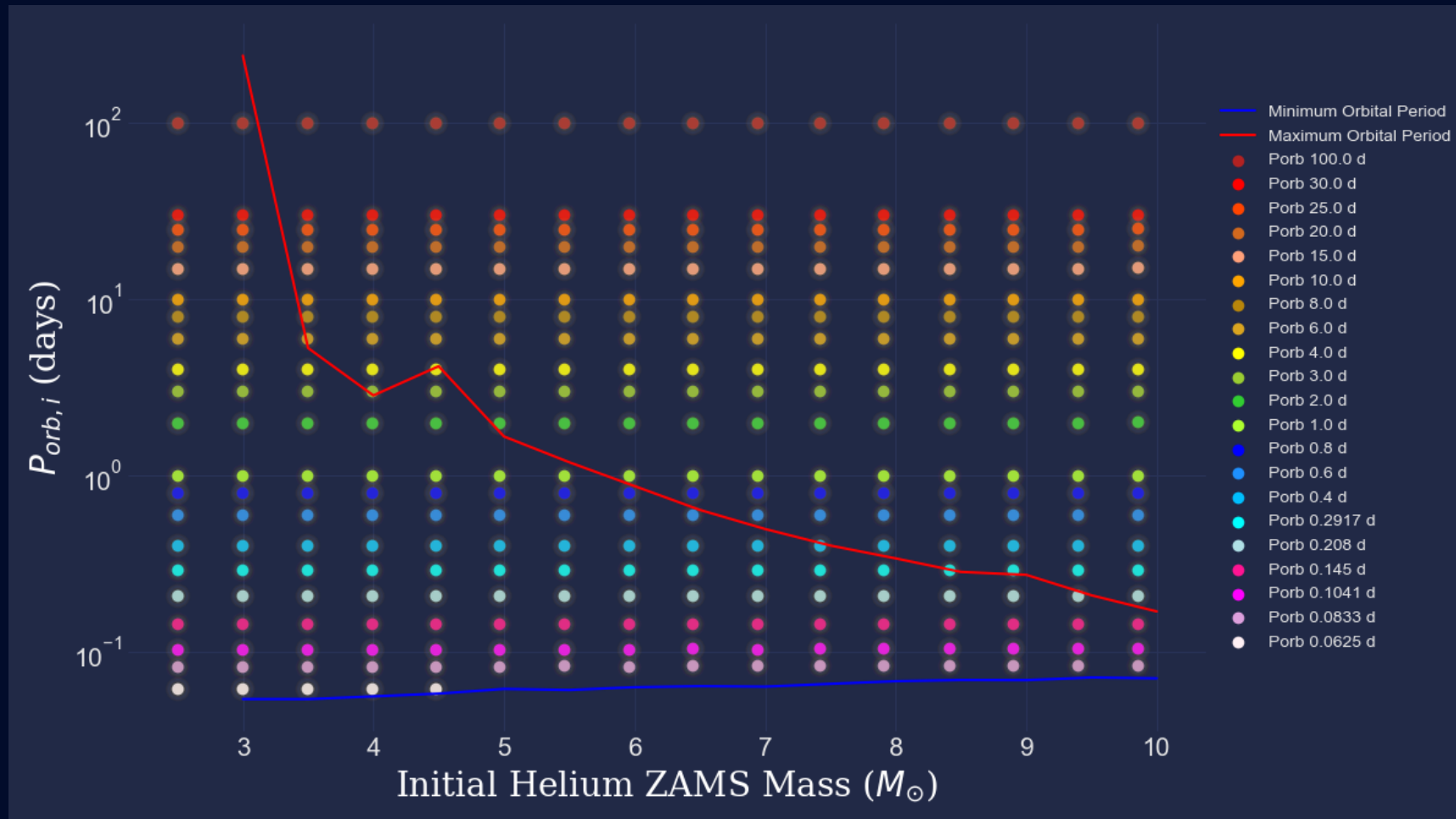
Peters 1964b
Mandel (2021b)

Initial and Maximum Radius as a Function of Initial He-ZAMS Mass at Solar Metallicity

(Recap)



Grid of Models Based On Initial He ZAMS (solar metallicity) Mass + $1.4 M_{\odot}$ NS and Initial Orbital Period

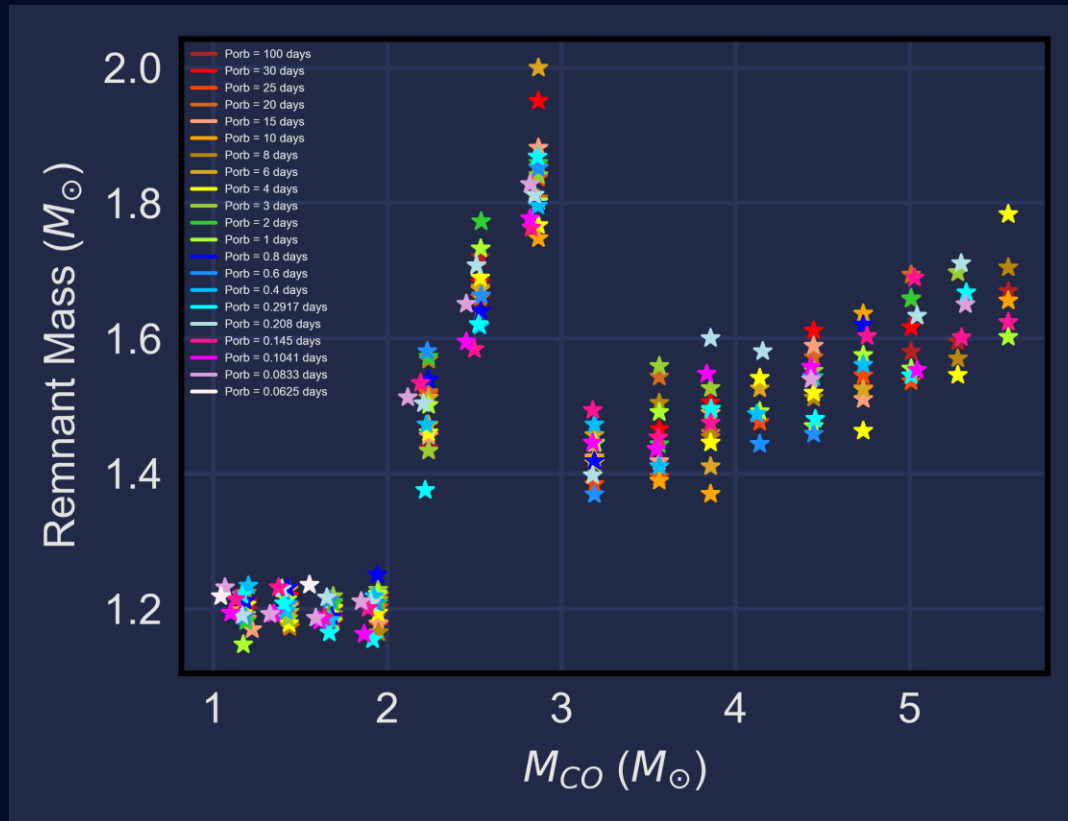




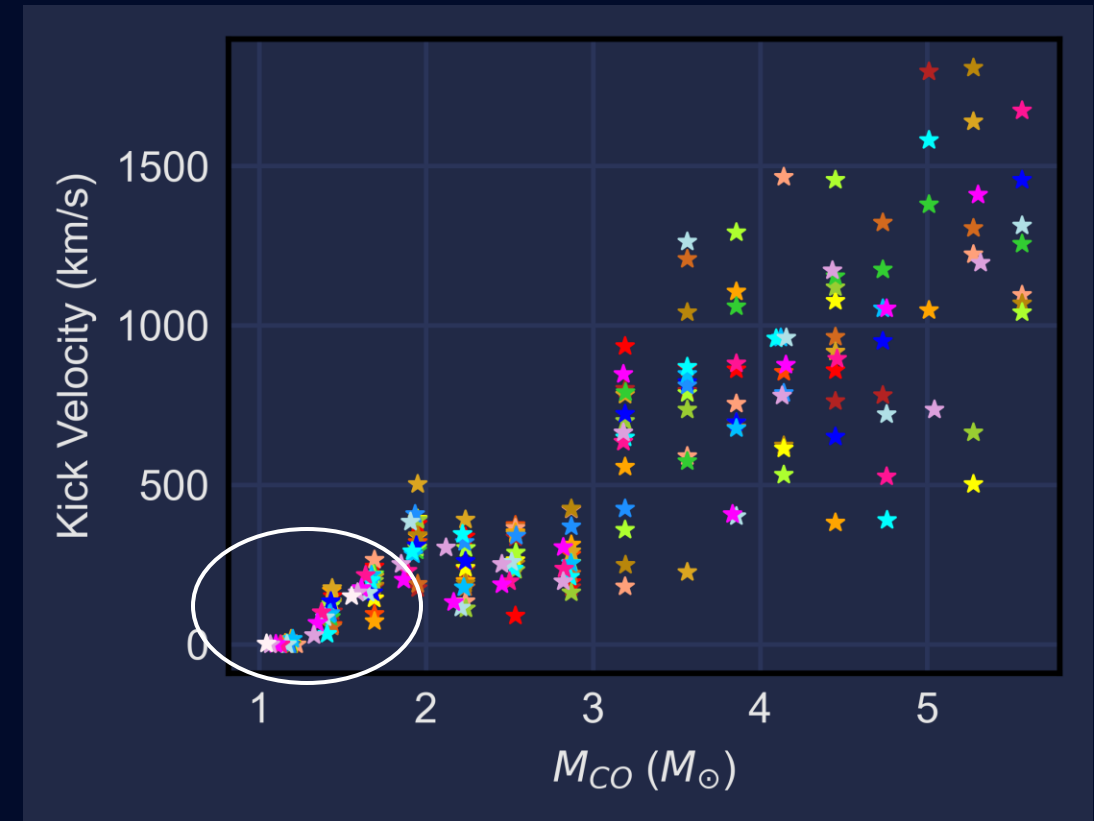
**Incorporating supernova and
kick prescription**

Mandel & Müller 2020,
Kapil et al. 2023

Standard Supernovae and Kick Prescription (Standard Model)



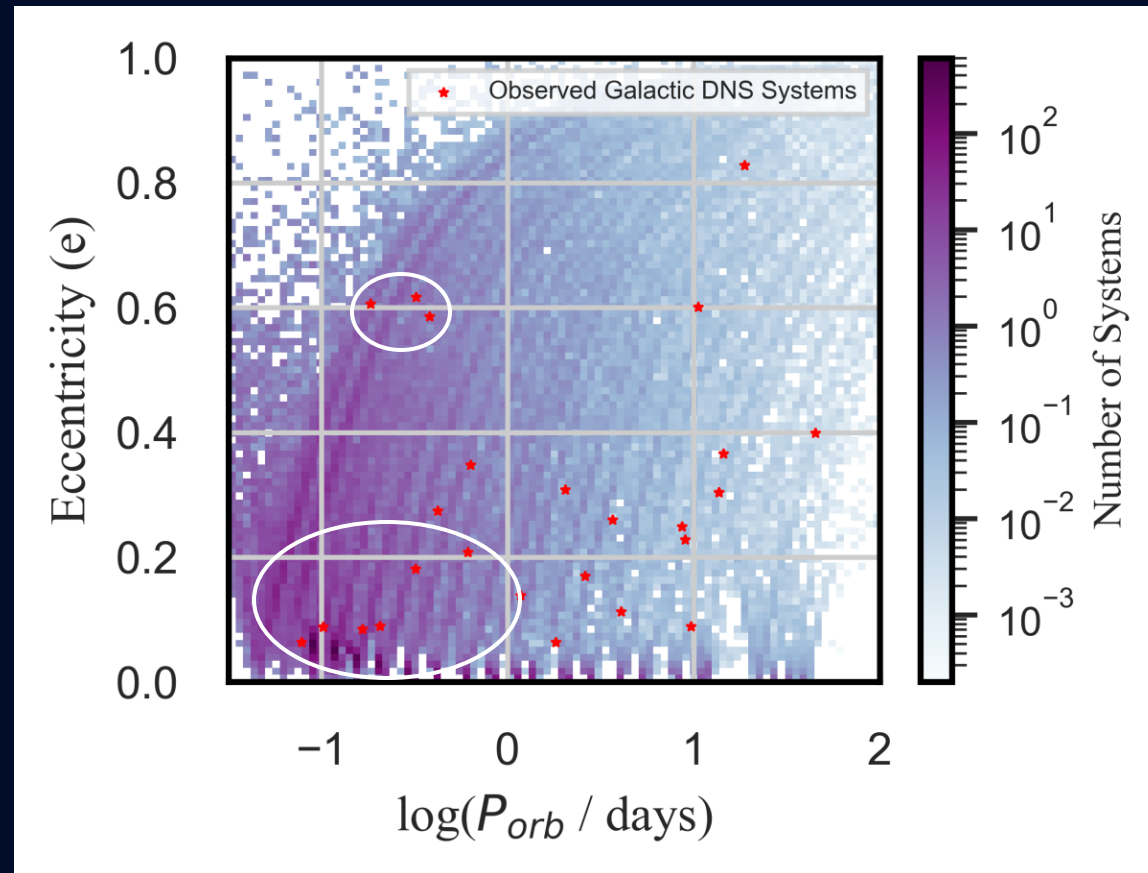
Remnant masses (NS2) as a function of CO core mass



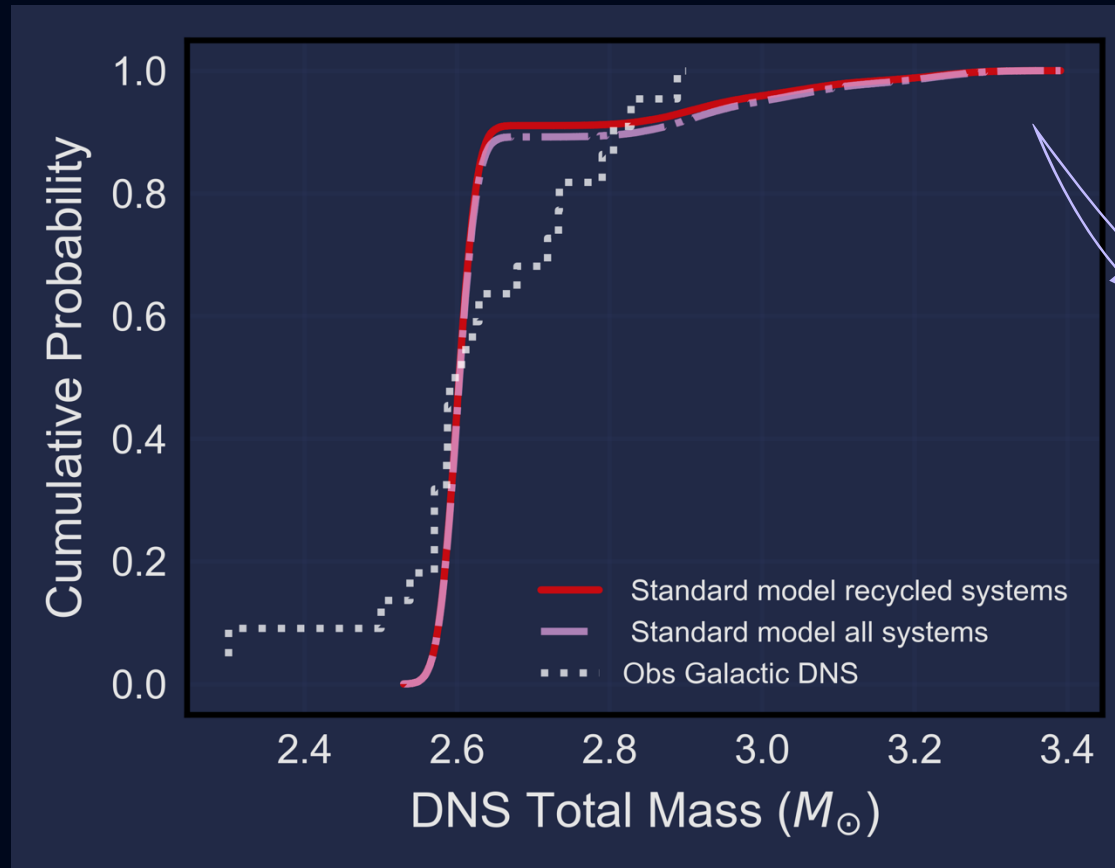
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Overproduce high mass systems
than observed

Assumed remnant mass and
CO core mass relation

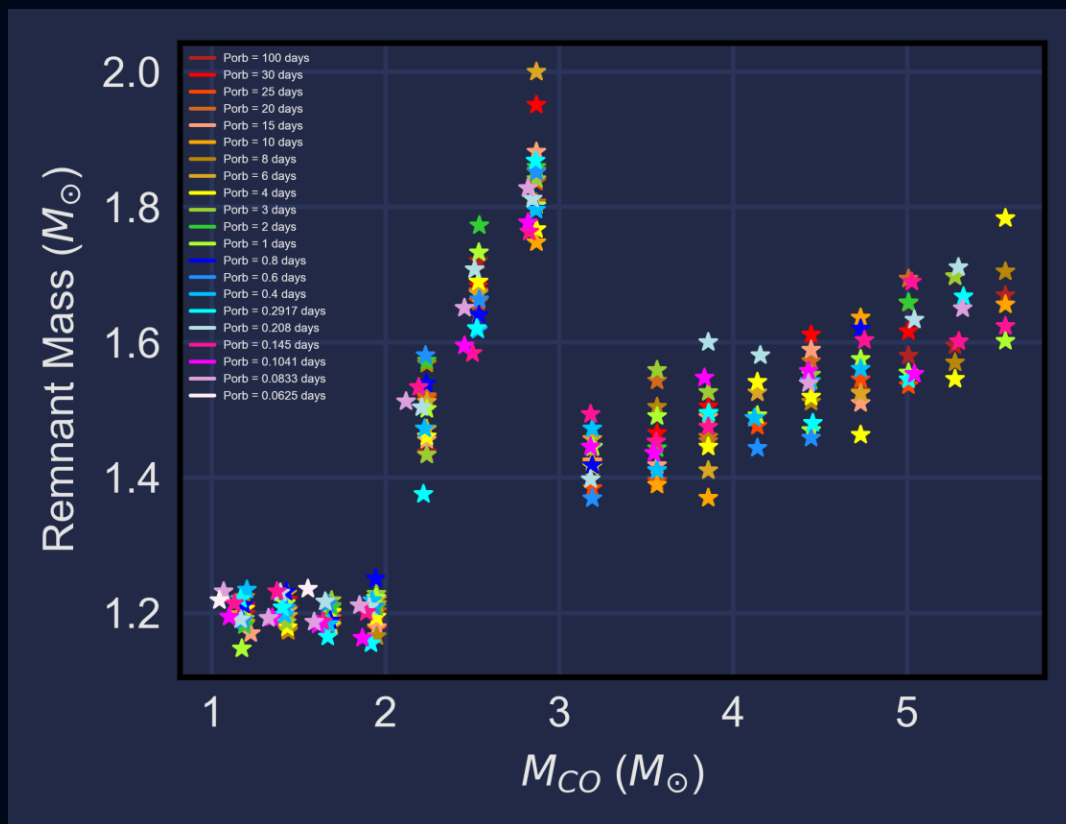
Issues with the Standard Prescription

- Overpredicts eccentricities
- Overpredicts DNS total mass than observed at solar metallicity

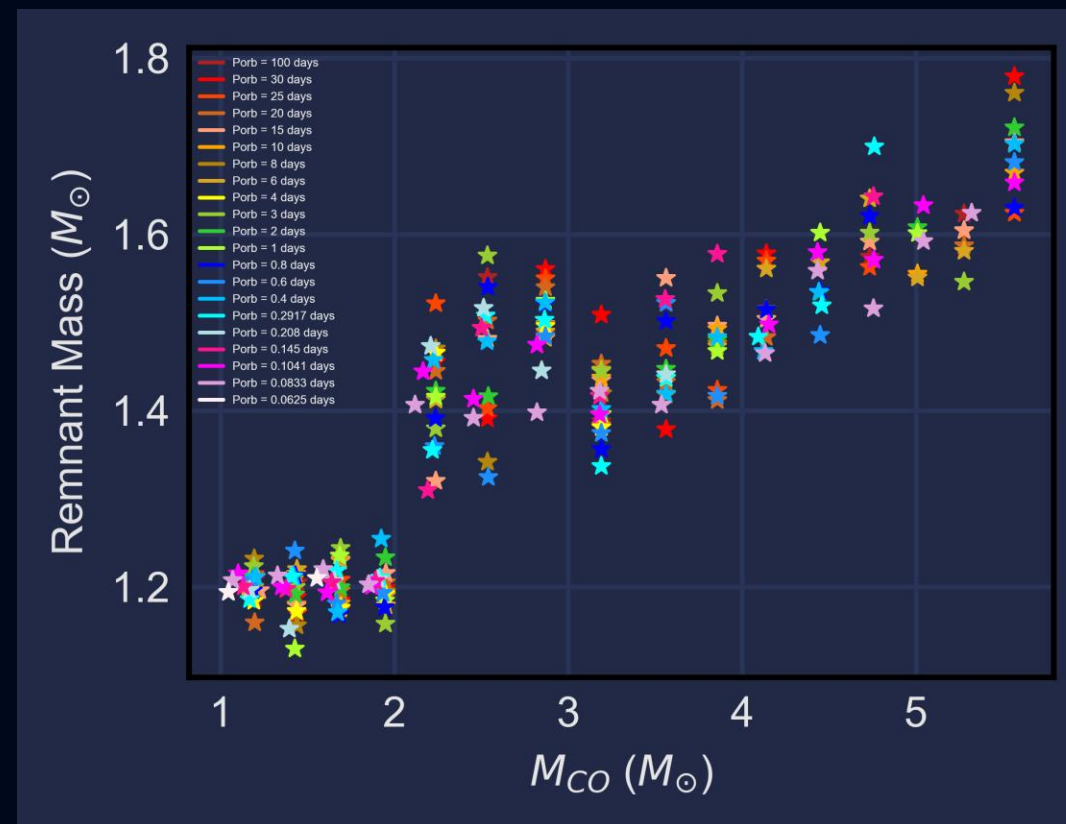
Modifications to the Standard Model

1. Different remnant mass and CO core mass relation
2. Incorporate Ultra-stripped and Electron capture supernovae conditions
3. Kick slow merging systems to high velocities

Standard Model (BEFORE)



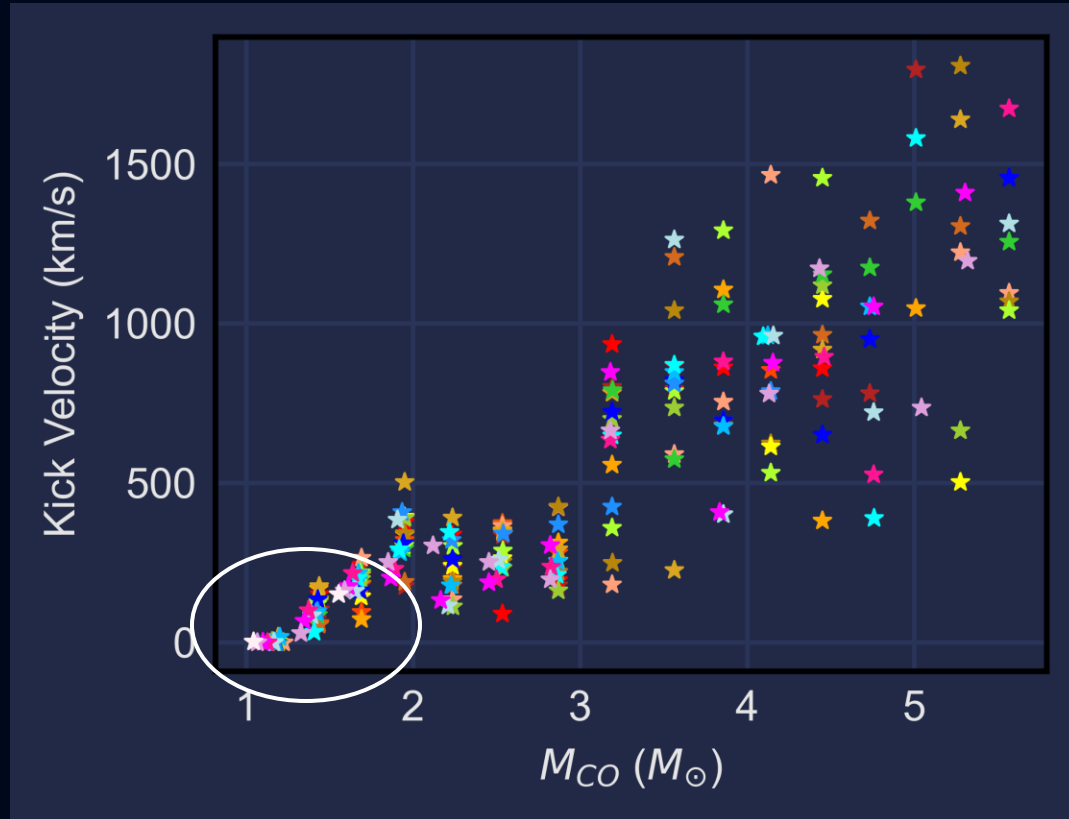
Modified Model



Remnant masses (NS2) as a function of CO core mass

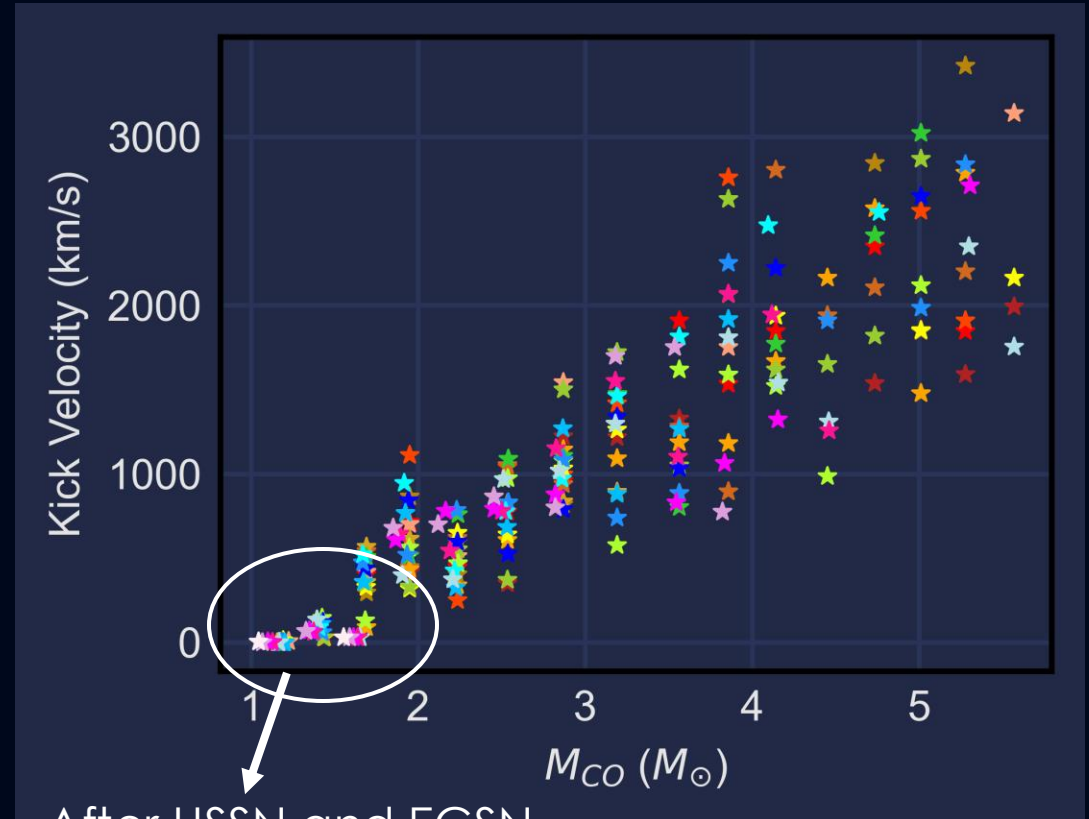
Standard Model

Before applying USSN and ECSN condition



Modified Model

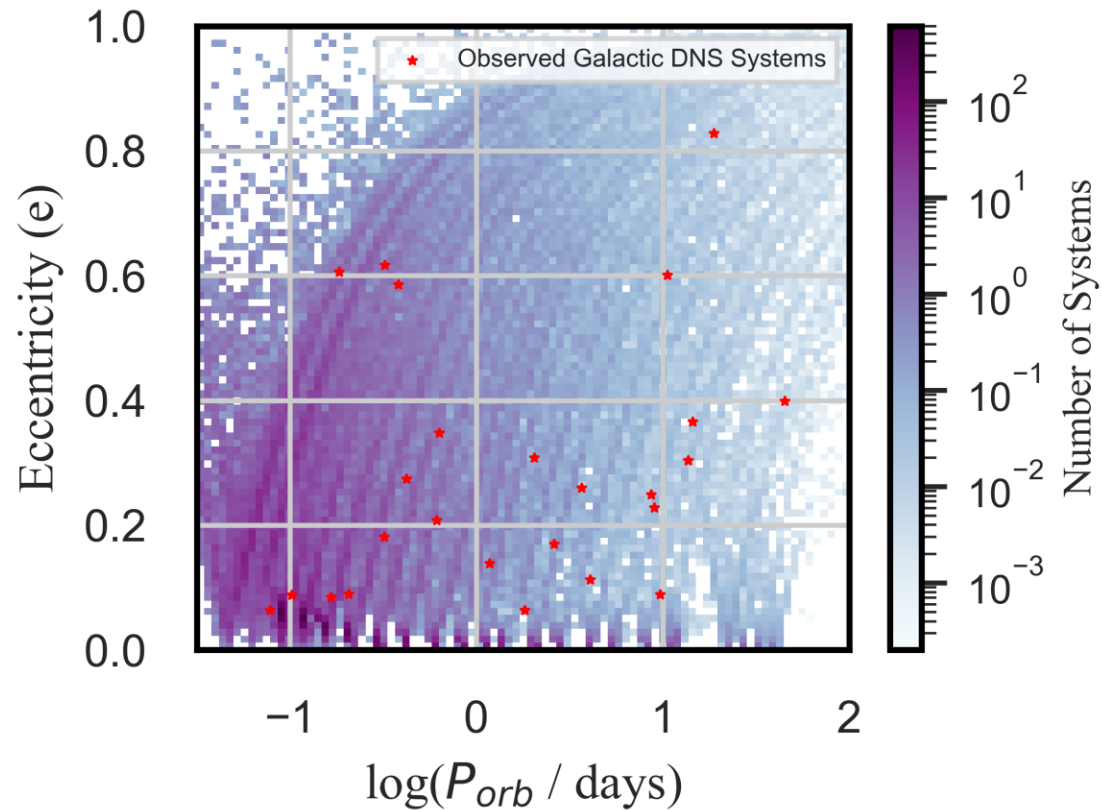
After applying USSN and ECSN condition



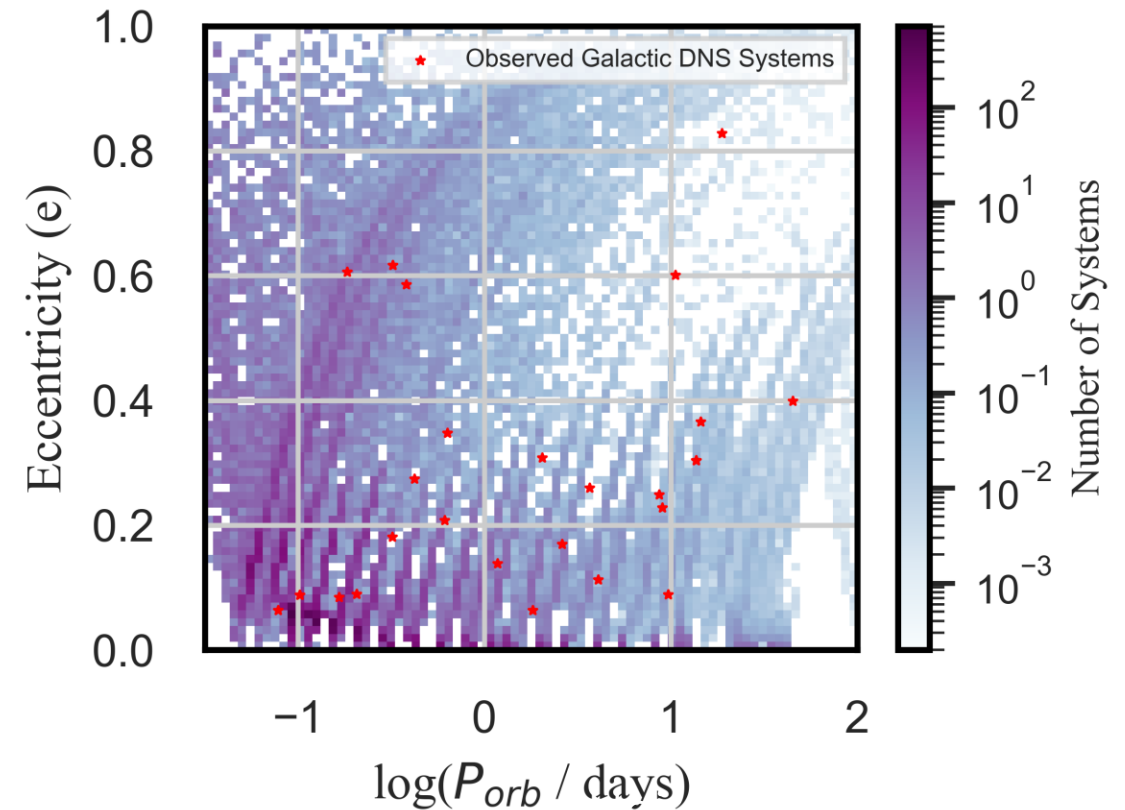
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined Porb-e Distribution of all models

Standard Model
(BEFORE)

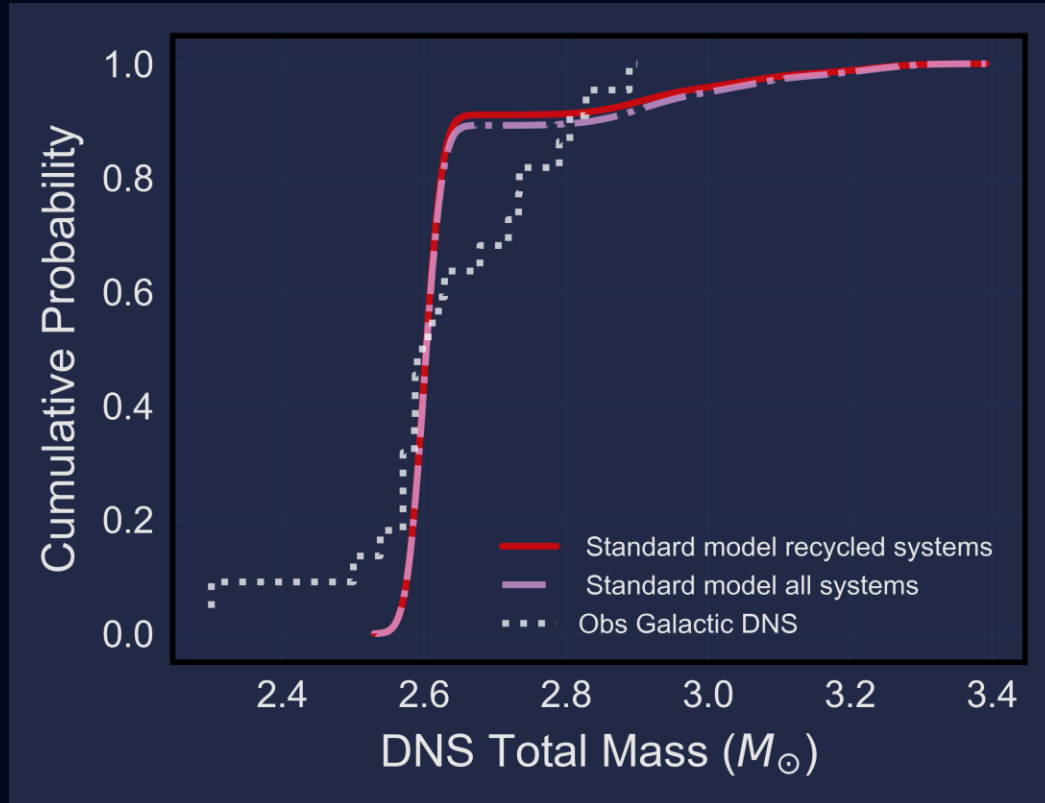


Modified Model

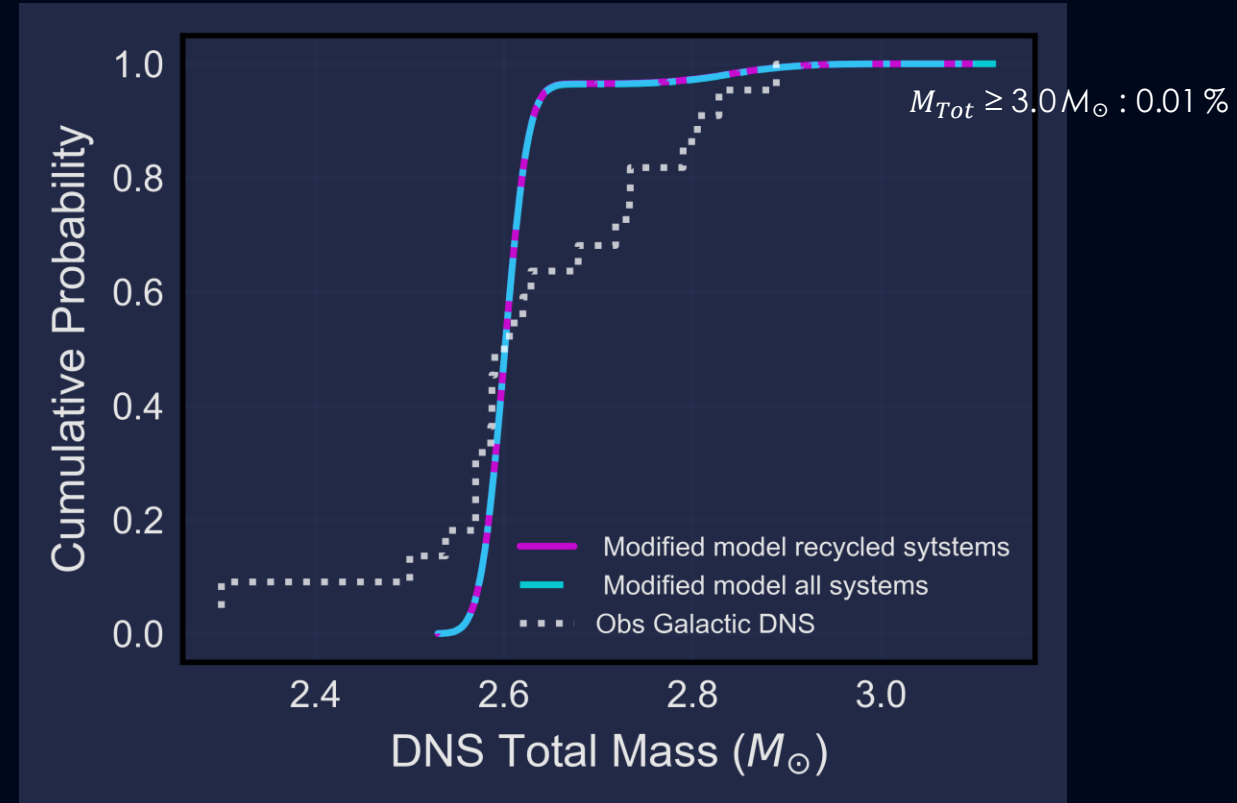


DNS Total Mass Distribution

Standard Model
(BEFORE)



Modified Model

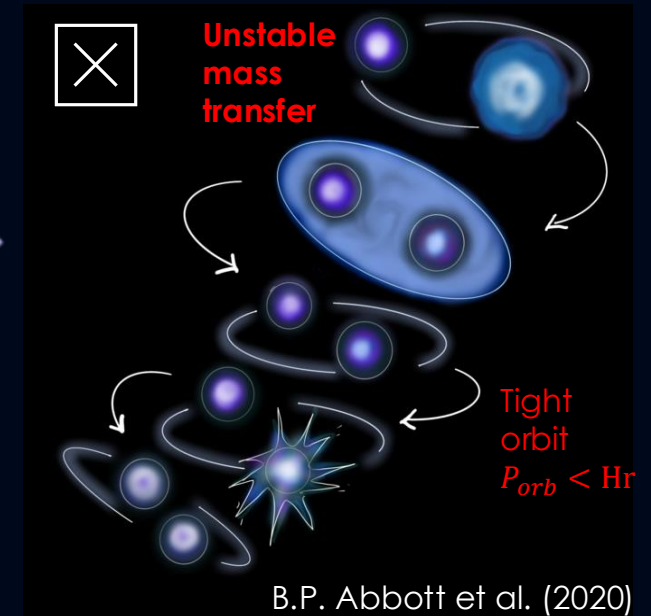
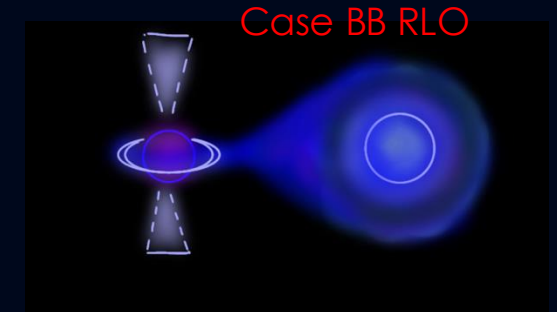


Nair et al. 2025 (In Prep)

GW190425 Progenitors

Modified: (Out of all the systems $\geq 3 M_{\odot}$)

- 82% systems - formed via Case BB RLO MT
- 18% systems - formed via massive helium stars without recycling the first-born NS
- Rules out the formation pathway via unstable mass transfer



Fewer DNS systems $\geq 3 M_{\odot}$

Project I, Part B

Explore

- *Low metallicities*
- *Massive first-born NS mass*
- *Super-Eddington accretion*

Paper I

Investigating the formation pathway of heavy double neutron stars: Part I

Ashwathi Nair,^{1,2★} Simon Stevenson,^{1,2}

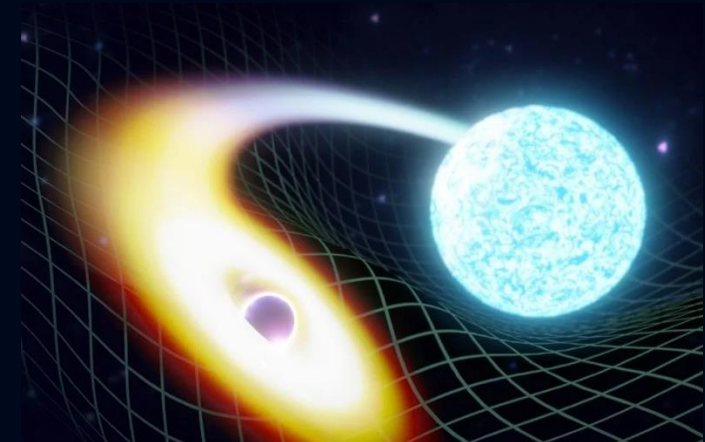
¹*OzGrav : The ARC Centre of Excellence for Gravitational Wave Discovery, Hawthorn, VIC 3122, Australia*

²*Centre for Astrophysics and Supercomputing, Swinburne University of Technology, Hawthorn, VIC 3122, Australia*

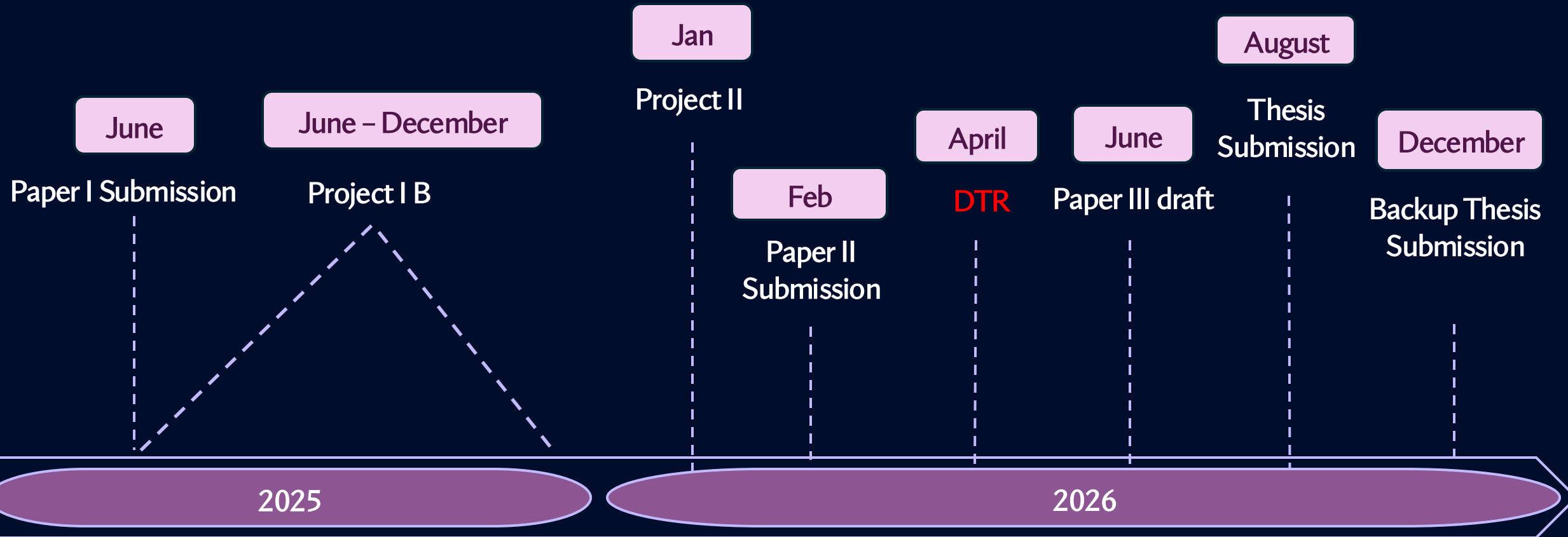
(To be submitted)

Project II

- Studying the formation pathway of any double neutron star (DNS) system detected during the ongoing LVK O4 run
- Analyzing the formation pathways and rates of simulated neutron star–black hole (NSBH) systems, and comparing them with all NSBH candidates detected via gravitational waves.



NSBH binary



Conclusion



Models calibrated to match the Galactic DNS population



Roughly reproduces the observed Galactic DNS distribution



Used models to explore **GW190425 progenitors**



Massive systems (like GW190425) formed at **solar metallicity** are **rare**



Unstable Case BB MT pathway does not yield GW190425-like binaries



Model limitations: Can't explain high GW190425 rates



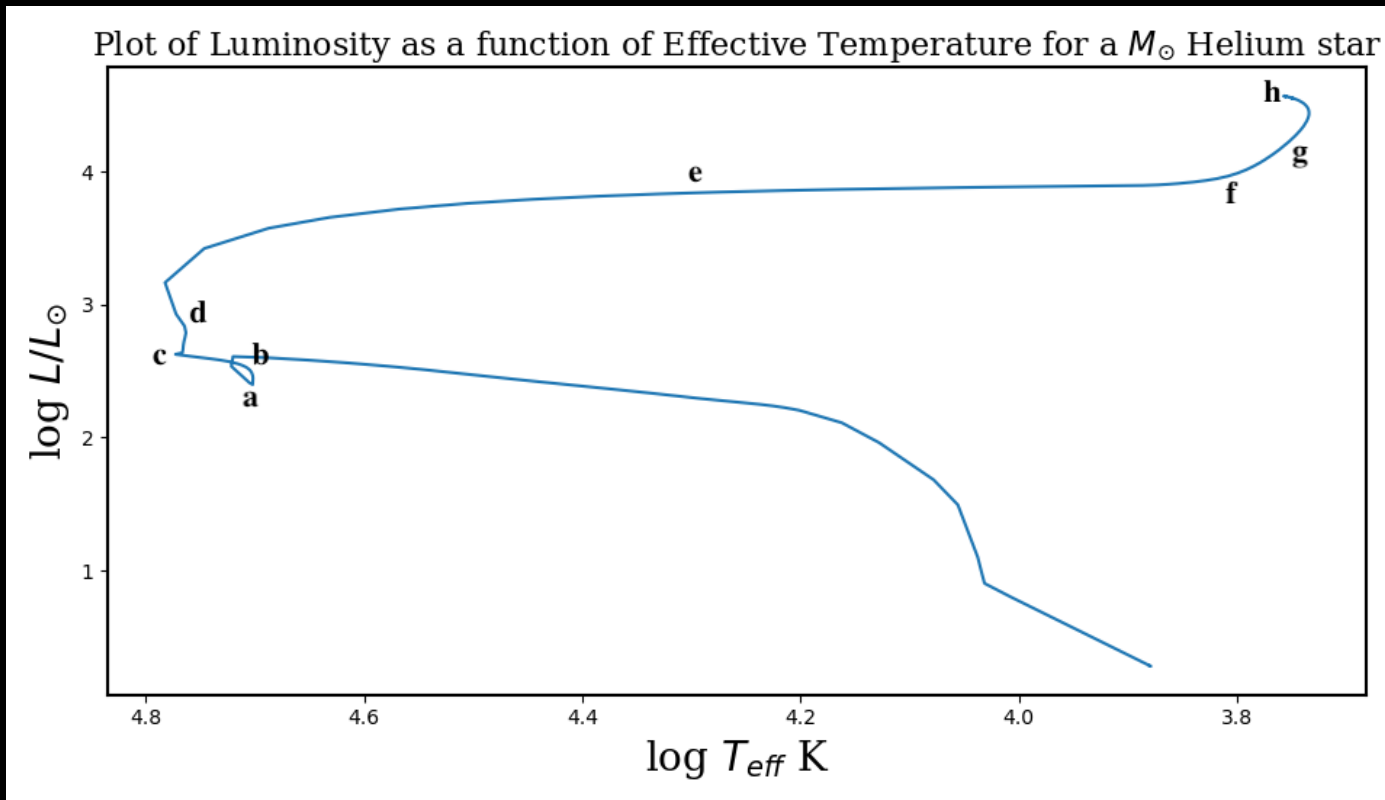
Next steps: Explore low-Z models, massive NS primaries & super-Eddington accretion



Hoping O4 run detects at least one DNS

EXTRA SLIDES

1 $M_{\odot}$ Helium Star Evolution



a : zero age main sequence

a - b : Main sequence phase

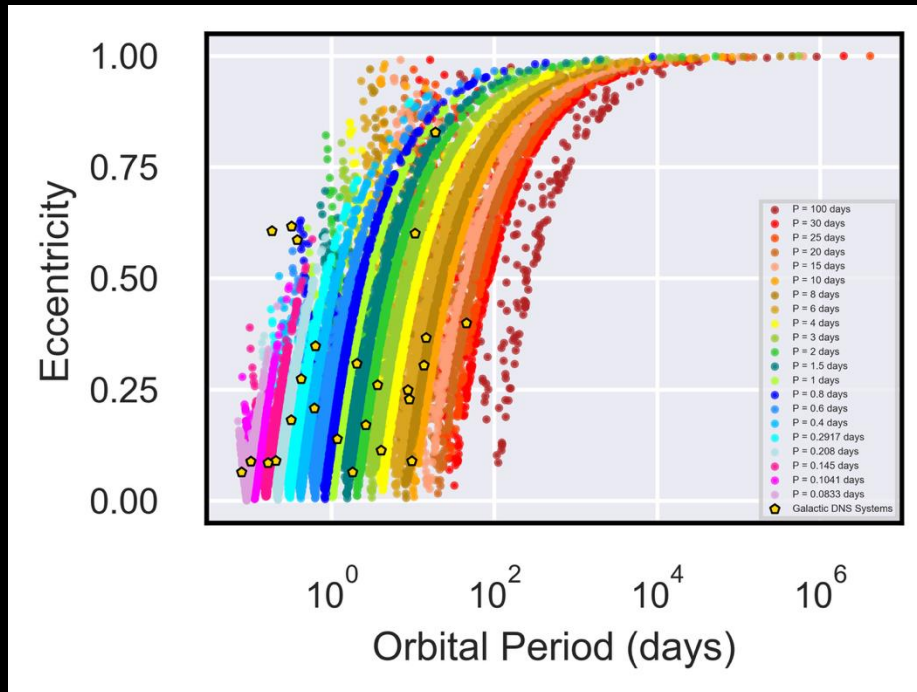
b - c : exhaustion of central helium,

c - d : first cooling phase,

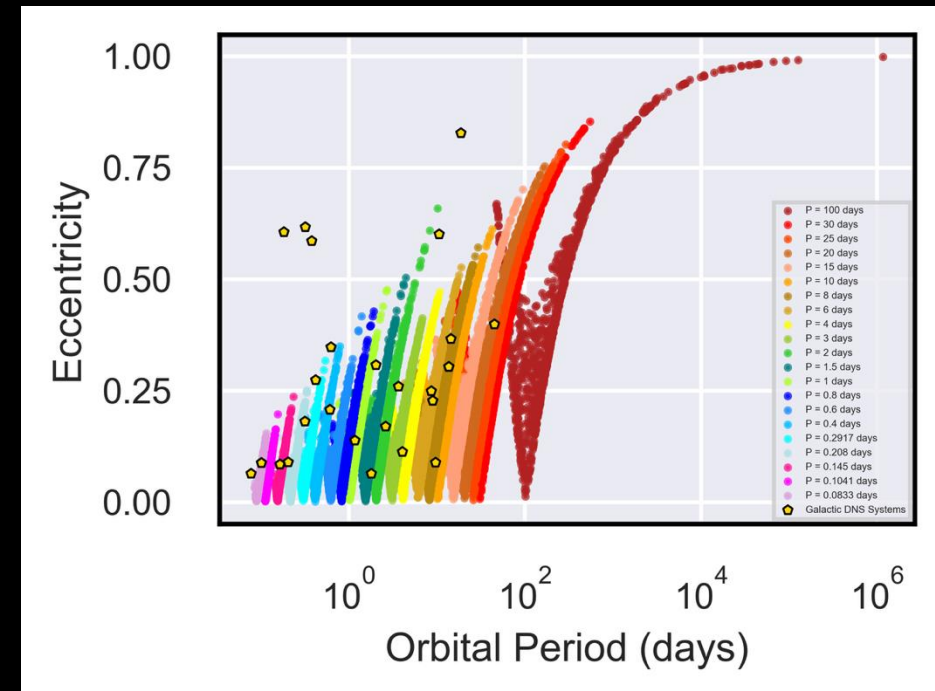
d - e : slow phase of shell helium burning

e - h : fast phase of helium shell burning

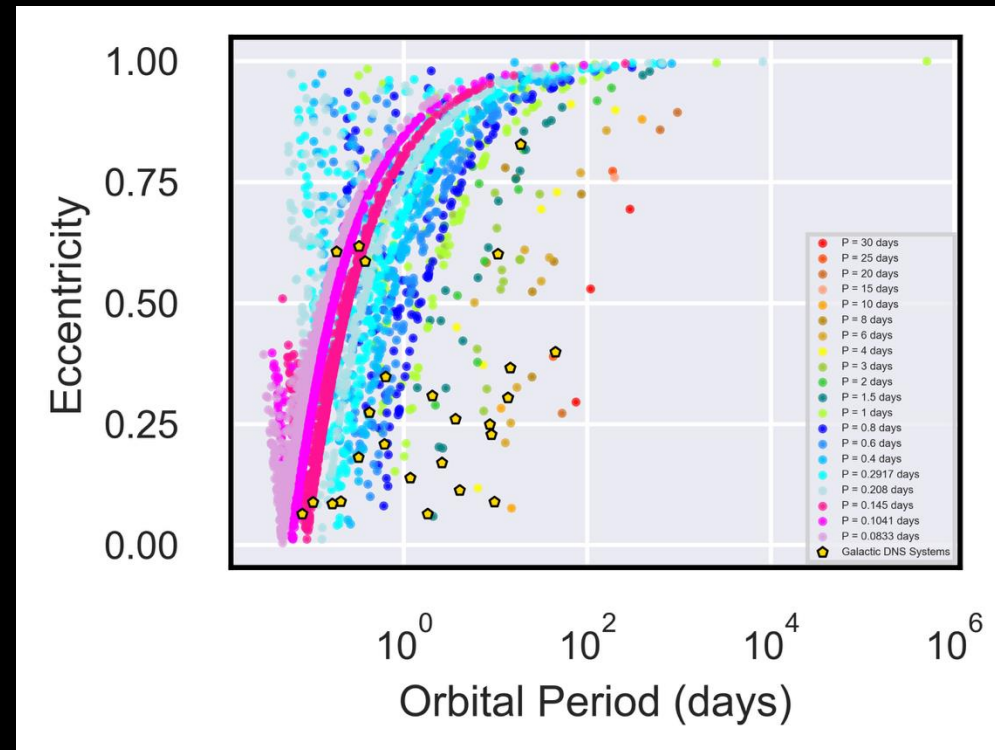
Standard model 3Msun



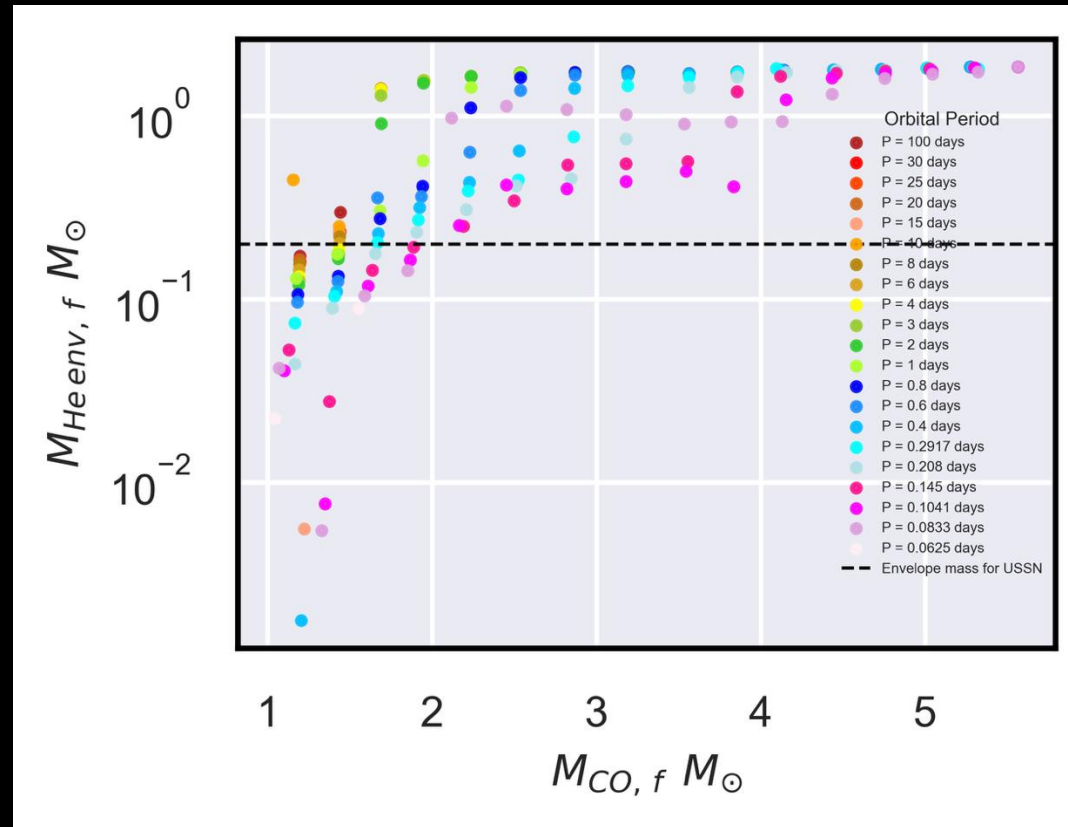
Modified 3Msun



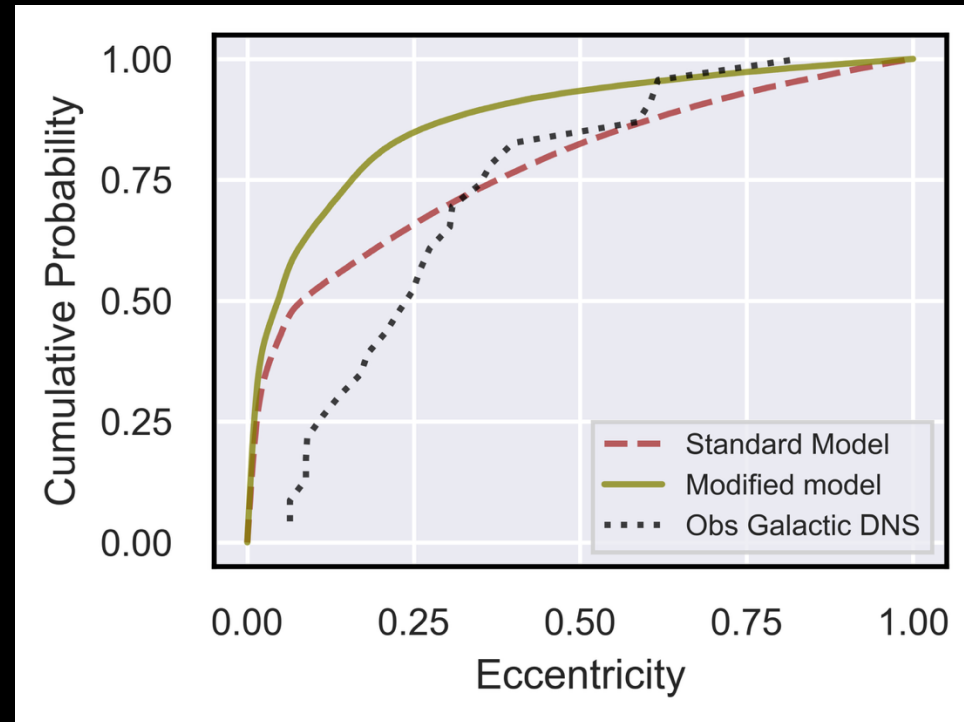
Modified 4Msun



Final helium envelope pre-SN



Eccentricity



Carbon-oxygen (CO) core masses as function of initial He-ZAMS mass

